Fuel For Thought

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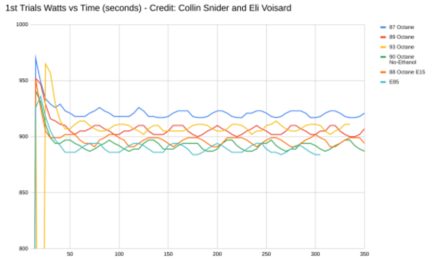
Scientific Question

All gasolines/fuels that you can fill your car with are very different in many ways. Most gasolines/ fuels begin to differ when you look at the concentrations of ethanol and octane values. We wanted to compare the efficiency under load of 87 octane, 89 octane, 93 octane, 90 octane no-ethanol, 88 octane E15, and E85. With this being said, we asked the question of what type of gasoline/ fuel can produce the most kilojoules of energy with 70 mL of liquid under different loads of powe

Figure 1 (Left) - The 6 different fuel cans
Figure 2 (Right) - Generator being used

Data Analysis & Results



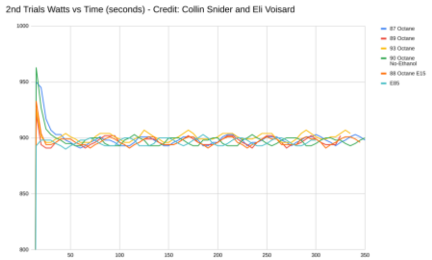
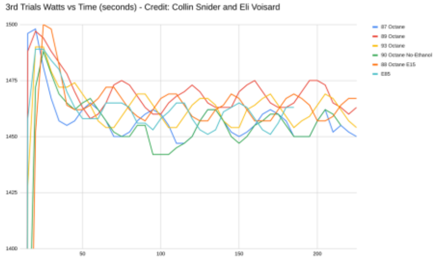


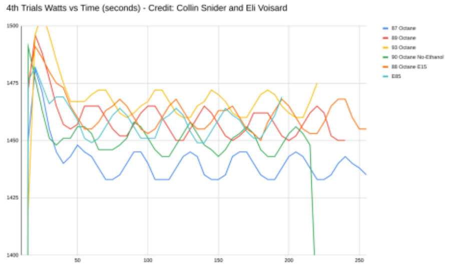
Figure 3 (top left)
Trial 1 of Watts vs. Time (sec) graph for 900 watt output over 350 seconds.

Figure 4 (bot. left)
Trial 2 of Watts vs. Time (sec) graph for 900 watt output over 350 seconds.

Figure 5 (top right)
Trial 3 of Watts vs. Time (sec) graph for 1500 watt output over 200 seconds.

Figure 6 (bot. right)
Trial 4 of Watts vs. Time (sec) graph for 1500 watt output over 200 seconds.





Methodology

- **Organization:** Fuels were separated into six cleaned, and independent gas cans (Figure 1). The carburetor bowl of the generator was emptied each trial to ensure no fuel was remaining in the tank (Figure 2).
- **Measuring equipment:** Our measuring equipment consisted of an inverter generator, 1500W max space heater and an arduino UNO with supporting interfaces. The generator we used was multi-fuel capable ensuring that some fuels with a higher octane rating such as E85 would not be at a disadvantage. The generator was able to accurately adjust to each fuel while running as it is equipped with electronic ignition and throttle control. Further, since it is an inverter generator, it was able to intelligently vary rpms to better match the load at hand, providing improved efficiency under different power loads. We chose to use a space heater as a constant load source since they are often found in industrial applications for generator testing. With this constant load we could measure the voltage and amperage to calculate wattage and incorporate time to calculate watt-hours or in our case kilojoules. To measure the voltage, we stepped down the 120 VAC to a about 9 VAC and then rectified and smoothed the voltage all using components from an old computer power supply. Then we could use a 100/330 voltage divider (shown in figure 7) to get the VDC within the 0-5VDC that can be measure by the Arduino's analog input. For our current measuring interface, we purchased a passive, hall-effect, current transducer that output a analog voltage 0-5VDC directly related to a reading of 0-20 amps. We programmed the arduino in such a way to (1) only take measurements when the current reading was greater than 2 amps so as to automatically start/ stop its internal timer & (2) take a readings of the values every 5 seconds which could be output into a serial monitor on our laptop and then a spreadsheet to make data compiling much easier after trails were completed.
- **Data:** Data was collected using the measuring equipment mentioned. These points were transferred from the Arduino to a spreadsheet in order to make the graphs (Figures 3, 4, 5, 6, 9, 10).
- **Comparison:** Data was compared in several different ways. First, by the consistency of watts produced. Second, by the amount of time the fuel was able to run the generator. Next, by the amount of kilojoules produced by the 70 mL of fuel. Lastly, by the average kilojoules of energy per dollar.

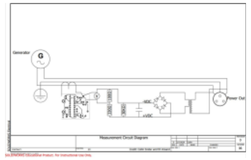
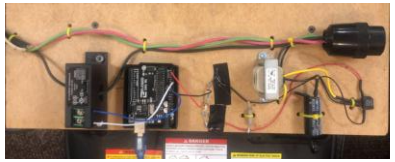


Figure 7 (left) - Electrical drawing of measuring equipment

Figure 8 (right) - measuring system used to collect data points.



Interpretation & Conclusion

- Eighty-eight octane E15 has been determined to be the most efficient fuel because it produces a high amount of kilojoules at a lower cost.
- On average, 87 octane produced the highest average amount of kilojoules, followed by 88 octane E15, 89 octane, 93 octane, 90 octane no ethanol, and E85 (Figure 9).
- We were able to use our own data, along with the average fuel prices in Ohio from the U.S. Department of Energy, to compare kilojoules of energy produced per dollar. Our graph (Figure 10) showed us that 88 octane E15 produced the most kilojoules of energy per dollar, followed by 87 octane, E85, 89 octane, 93 octane, and 90 octane no ethanol.
- The time of year we tested (mid January) may have an affect on the results, as fuel additive concentrations change depending on climate.
- The concept of testing and comparing fuels has certainly been done before. We believe our experiment is different than most other experiments because we tested the fuels in a controlled environment with a internal combustion engine rather than comparing theoretical energy content or efficiency in a moving car which could be influenced by outside factors such as wind or grade of the road.
- Using 88 octane E15 may be something that some drivers will begin to fill up there cars with because it is cheap while producing high amounts of kilojoules. This experiment also proved that more premium fuels do not perform better, but may be able to keep your engine cleaner and healthier, however we did not test this.

Figure 9 (left) - Graph of kilojoules of energy produced in each trial and the averages.
Figure 10 (right) - Graph of kilojoules of energy produced per dollar.